SHIVANI T M

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SUMMARY

A PhD researcher specializing in food technology, microbiology, and probiotic science. Experienced in probiotic isolation, characterization, and bioactivity profiling for functional food applications. Skilled in microbial fermentation, biofilm analysis, antimicrobial resistance testing, cell line assays and metabolic profiling. Proficient in molecular biology techniques, including RNA detection, electrophoresis, and bioinformatics for microbial analysis. Experienced in food safety, quality control, and enzyme technology. Passionate about developing innovative probiotic-based products and applying biotechnology for enhanced food functionality and health benefits.

EXPERIENCE

TEACHING AND RESEARCH ASSISTANT, 09/2020 - Current

Vellore Institute of Technology, Vellore, India

- Researching on **probiotics**, focusing on their **isolation, characterization, bioactivity, and metabolite analysis**, with an emphasis on novel strains from unconventional sources.
- Integrating **microbial metabolomics and functional analysis** to explore probiotic mechanisms and their applications in **non-dairy fermented health beverages.**
- Assisting in laboratory sessions and mentoring students, fostering scientific curiosity and research skills in microbiology and biotechnology.
- Conducted laboratory sessions in **Basic Microbiology, Industrial Biotechnology, Food and Soil Microbiology, Cell Biology, and J Components.**
- Developed and implemented **new course materials**, creating a structured teaching plan to enhance student engagement and practical learning.
- Bridging research and education by incorporating **real-world microbiological applications into teaching**, enabling students to understand the translational impact of microbial studies.

RESEARCH ASSISTANT, 06/2020 - 08/2020

BITS PILANI, Hyderabad Campus, Hyderabad, India

- **Worked on developing a single RNA detection method** using an existing platform, enhancing **sensitivity and accuracy.**
- Optimized **electrophoresis, agarose gel preparation, and protocols** to improve detection efficiency.
- **Reduced background noise and improved signal clarity**, making RNA analysis more reliable.
- Provided a **cost-effective and accessible** method for **molecular diagnostics, food safety, and research applications.**
- Potential applications in **food microbiology**, including **pathogen detection, fermentation monitoring, and probiotic gene expression studies.**
- Strengthened expertise in **molecular biology techniques and analytical methods** for RNA analysis.

INTERN, 05/2019 - 06/2019

Bio-Axis DNA Research Centre, Hyderabad, India

- Isolated pectinase-producing fungi, and optimized enzyme production through submerged fermentation.
- **Enhanced apple juice yield** by improving pectin breakdown efficiency.
- **Immobilized pectinase in calcium alginate beads**, increasing **stability and reusability.**
- **Provided a cost-effective, eco-friendly alternative** to commercial enzymes for food processing.
- Strengthened skills in **microbiology, enzyme technology, and bioprocess optimization.**

PUBLICATIONS

Probiotic evaluation, adherence capability and safety assessment of Lactococcus lactis strain isolated from an important herb "Murraya koenigii"
TM Shivani, M Sathiavelu
Scientific Reports 14 (1), 19769

A comprehensive review on functionality of probiotics in edible packaging
TM Shivani, M Sathiavelu
Packaging Technology and Science 36 (1), 15-30

Role of bacteria and fungi in the circular agriculture economy
L Pillai, JS Saravanan, TM Shivani, S Sur, M Sathiavelu
The Potential of Microbes for a Circular Economy, 131-148

ONGOING PROJECT WITH COLLABORATION WITH ICAR-IIHR

Ongoing Project Collaboration: Development of Probiotic Ambali Premix Using Millets and Encapsulation Technology

Collaborating on the formulation of a probiotic ambali premix, leveraging millets as a nutrient-dense, gluten-free base to support the growth and stability of *Lactococcus lactis*.

- **Developing a unique probiotic ambali premix** using **millets** for a gluten-free, nutrient-rich base.
- **Enhancing probiotic stability** with **advanced encapsulation and lyophilization technology** for better shelf life.
- **Optimizing fermentation** to retain **bioactive compounds** and maximize health benefits.
- **Creating a non-dairy, functional beverage**, addressing **lactose intolerance and vegan needs.**
- **Bridging traditional fermentation with modern food tech** for a **sustainable, accessible solution**.

WORKSHOPS AND CONFERENCES

- MLT(Medical Lab Technician) course in Ethiraj college for women
- Workshop on bioinformatics at Presidency College, Chennai
- Workshop on Chromatographic Techniques, Sathyabhama University
- Workshop on Human Microbiome, Mizoram University
- One-day International Conference on International Microorganisms Day (IMD) in association with FEMS (Federation of European Microbiological Societies)
- Three-day course on Techniques for 'Genetic Manipulation in Bacteria and Viruses' Conducted by the School of Biology, MIT World Peace University, India
- Participated in International Conference on Emerging Trends in Biotechnology (ICTEB), VIT, Vellore

EDUCATION AND TRAINING

Vellore Institute of Technology, *Vellore, Tamil Nadu, 2020*

Ph.D.: Microbial Biotechnology And Probiotics

- My PhD research focuses on the isolation, characterization, and application of probiotic bacteria from unconventional sources, investigating their health benefits and bioactive metabolite production. I employ an interdisciplinary approach combining microbiological techniques, biochemical analysis, and bioinformatics tools, including metabolite profiling and docking studies, to explore the therapeutic potential of these probiotics. My work also involves hands-on practice with advanced techniques such as microscopy, spectroscopy, and molecular biology methods, enhancing my ability to analyze and interpret complex data. The ultimate goal is to develop non-dairy probiotic formulations while contributing to a better understanding of probiotic functionality and metabolic pathways, with a focus on practical applications in the food, health, and wellness industries. Through my research, I aim to make significant contributions to the field of probiotics, improving public health and expanding the potential of microbial biotechnologies.

***Ethiraj College For Women**, Chennai, Tamil Nadu, 2020*

Master of Science

- 8.2 GPA

***SKR& SKR Women's Degree College**, Kadapa. Andhra Pradesh, 2018*

Bachelor of Science: Biochemistry, Microbiology, Chemistry

- 9.15 GPA

***Narayana Junior College**, Kadapa, Andhra Pradesh, 2015*

Intermediate

- 92.7%

***Army Public School**, Nagrota, Jammu & Kashmir, 2013*

Matriculation

- 8.0 GPA

SKILLS

Software skills

- MS Office
- GraphPad Prism
- ChemDraw
- Origin
- EndNote
- Mendeley
- MEGA
- SPSS

Bioinformatics tools

- **ResFinder** (Antimicrobial Resistance Finder)
- **VirulenceFinder** (Virulence Factor Finder)
- **RAST** (Rapid Annotations using Subsystems Technology)
- **CARD** (Comprehensive Antibiotic Resistance Database)
- **KEGG** (Kyoto Encyclopedia of Genes and Genomes)
- **MetaCyc** (Metabolic Pathways Database)
- **Microbial Genome Database** (Microbial Genomics and Metagenomics Database)
- **PlasmidFinder** (Plasmid Detection Tool)
- **Prokka** (Prokaryotic Genome Annotation)
- **BLAST** (Basic Local Alignment Search Tool)
- **NCBI GenBank** (Sequence Database)
- **GeneBank** (Genetic Sequence Database)

Food Technology and Microbiology Skills

- **Fermentation and Food Microbiology:** Microbial fermentation, probiotic production, microbial identification, biofilm analysis, and antimicrobial resistance testing
- **Metabolite Profiling & Bioactivity**: Extraction, identification, and bioactivity analysis of bioactive metabolites.
- **Food Safety & Quality Control**: GMP, HACCP, regulatory compliance.
- **Food Processing & Engineering**: Food preservation, thermal processes, packaging.
- **Enzyme Technology & Nutritional Analysis**: Enzyme application, food composition, nutritional profiling.
- **Shelf-life & Packaging Technology**: Stability testing, packaging methods.

Instruments handling

- FTIR
- GCMS
- HPLC
- Freeze drying—lyophilization
- Encapsulation & Targeted Delivery Systems
- Scanning Electron Microscopy (SEM)
- Confocal Laser Scanning Microscopy (CLSM)
- Spectrophotometer (UV-Vis and Fluorescence)
- Rheometer & Viscometer

LANGUAGES

Telugu: First Language

English: C2

Proficient (C2)

Hindi: C2

Proficient (C2)

Tamil: A1

Beginner